Chapter 02: Cloud Computing Architecture

By: Diwakar Upadhyaya

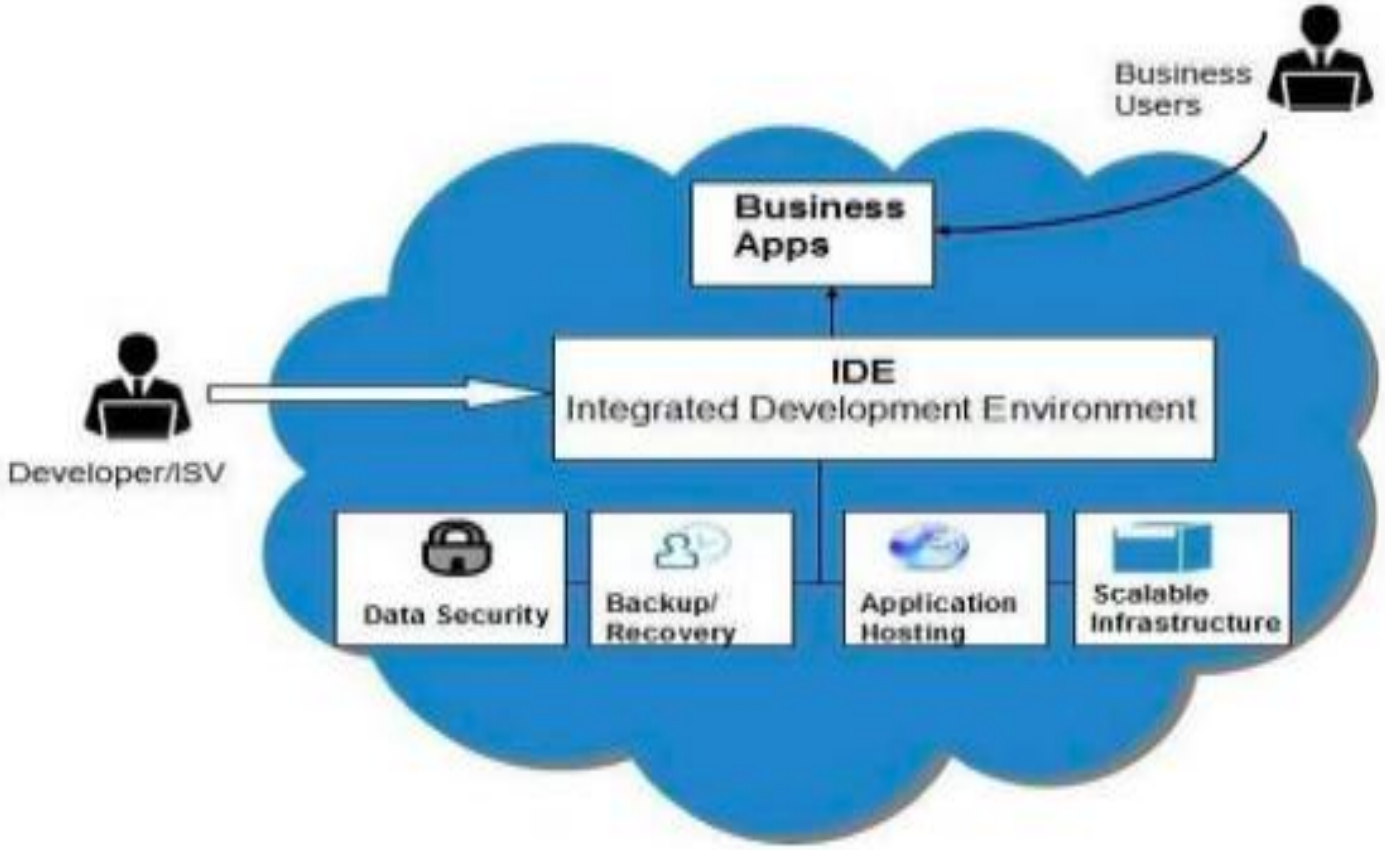
Platform-as-a-Service (PaaS):

- PaaS offerings include workflow facilities for application design, application development, testing, deployment, and hosting, as well as application services such as virtual offices, team collaboration, database integration, security, scalability, storage, persistence, state management, dashboard instrumentation, etc.
- The PaaS model makes all of the facilities required to support the complete life cycle of building and delivering web applications and services entirely available from the Internet, all with no software downloads or installation for developers, IT managers, or end users

- PaaS developers are concerned only with web-based development and generally do not care what operating system is used
- PaaS provides the capability for consumers to have applications deployed without the burden and cost of buying and managing the hardware and software
- PaaS facilitates immediate business requirements such as application design, development and testing at a fraction of the normal cost.

The Traditional On-Premises Mode

- The traditional approach of building and running on-premises applications has always been complex, expensive, and risky.
- Building own solution has never offered any guarantee of success.
- Each application was designed to meet specific business requirements.
- Each solution required a specific set of hardware, an operating system, a database, often a middleware package, email and web servers, etc.

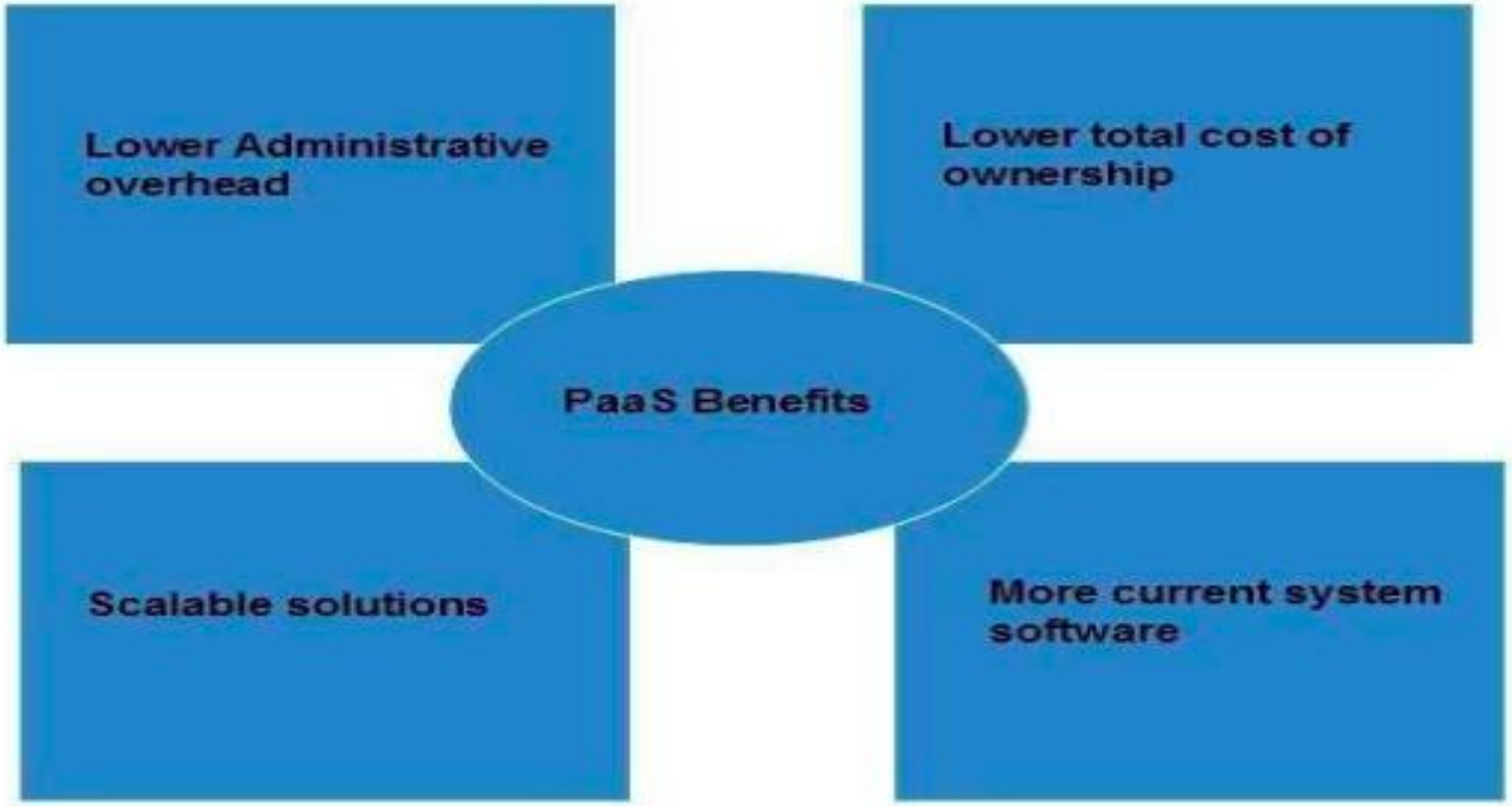

Business Users
Business Apps
IDE
Integrated Development Environment
Developer/ISV
Data Security
Backup/ Recovery
Application Hosting
Scalable Infrastructure

Key Characteristics of PaaS

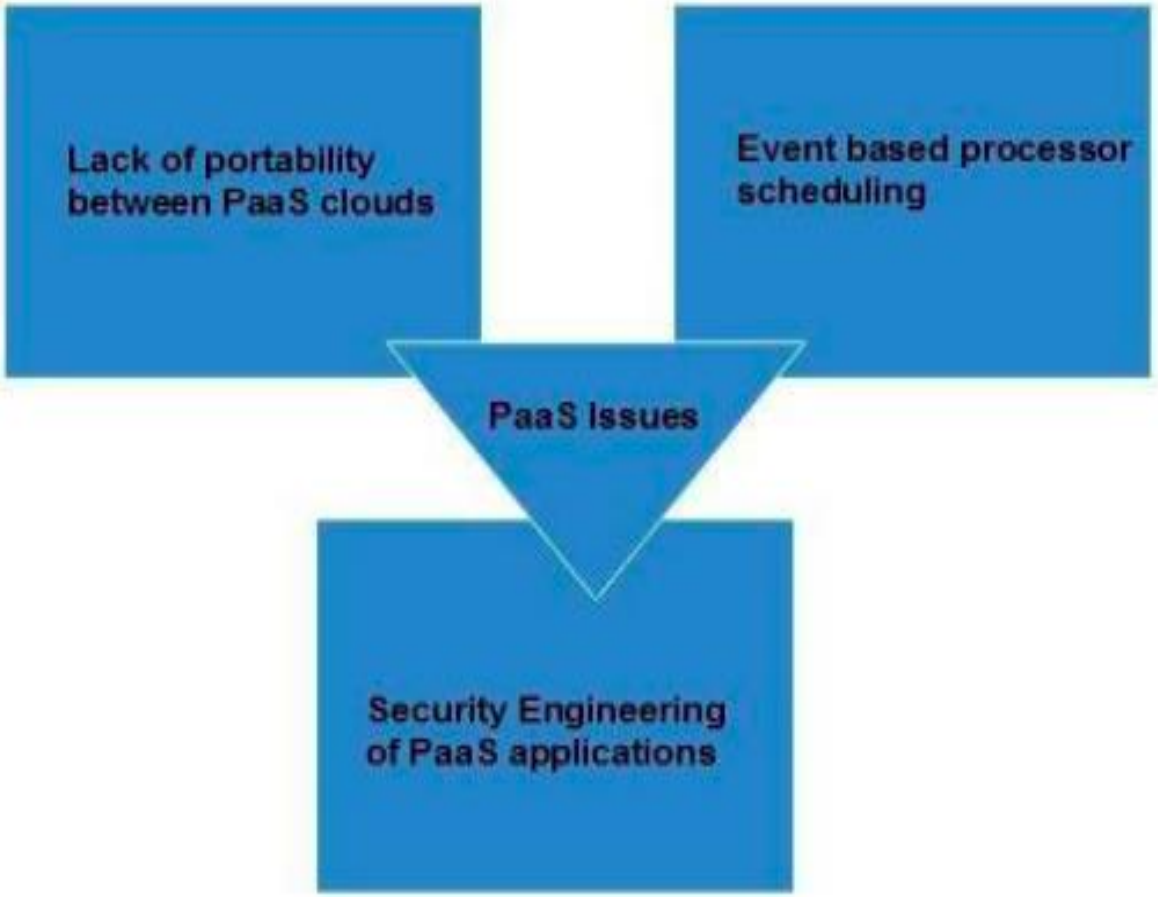
- Chief characteristics of PaaS include services to develop, test, deploy, host, and manage applications to support the application development life cycle
- Supporting a multitenant architecture helps to remove developer concerns regarding the use of the application by many concurrent users.
- PaaS providers often include services for concurrency management, scalability, fail-over and security.
- Another characteristic is the integration with web services and databases

- The ability to form and share code with ad-hoc, predefined, or distributed teams greatly enhance the productivity of PaaS offerings
- Integrated PaaS offerings provide an opportunity for developers to have much greater insight into the inner workings of their applications and the behavior of their users by implementing dashboard- like tools to view the inner workings based on measurements such as performance, number of concurrent accesses, etc.

Benefits:


Lower Administrative overhead
Lower total cost of ownership
PaaS Benefits
Scalable solutions
More current system software

Issues with PaaS



1. Lack of portability between PaaS clouds

- Although standard languages are used, yet the implementations of platform services may vary.
- For example, file, queue, or hash table interfaces of one platform may differ from another, making it difficult to transfer the workloads from one platform to another.

2. Event based processor scheduling

- The PaaS applications are event-oriented which poses resource constraints on applications, i.e., they must answer a request in each interval of time.

3. Security engineering of PaaS applications

- Since PaaS applications are dependent on network, they must explicitly use cryptography and manage security exposures.

PaaS Types:

1. Stand-alone development environments

- The stand-alone PaaS works as an independent entity for a specific function. It does not include licensing or technical dependencies on specific SaaS applications.

2. Application delivery-only environments

- The application delivery PaaS includes on-demand scaling and application security.

3. Open platform as a service

- Open PaaS offers an open-source software that helps a PaaS provider to run applications.

4. Add-on development facilities

- The add-on PaaS allows to customize the existing SaaS platform.

Software-as-a-Service(SaaS)

- Software-as–a-Service (SaaS) model allows to provide software application as a service to the end users.
- It refers to a software that is deployed on a host service and is accessible via Internet.
- There are several SaaS applications listed below:

Billing and invoicing system

Customer Relationship Management (CRM) applications

Help desk applications

Human Resource (HR) solutions

Characteristics of SaaS are:

- SaaS makes the software available over the Internet.
- The software applications are maintained by the vendor.
- The license to the software may be subscription based or usage based. And it is billed on recurring basis.
- SaaS applications are cost-effective since they do not require any maintenance at end user side.
- They are available on demand.

- They can be scaled up or down on demand.
- They are automatically upgraded and updated.
- SaaS offers shared data model. Therefore, multiple users can share single instance of infrastructure. It is not required to hard code the functionality for individual users.
- All users run the same version of the software.

Benefits

- Using SaaS has proved to be beneficial in terms of scalability, efficiency and performance.
- Modest software tools
- Efficient use of software licenses
- Centralized management and data
- Platform responsibilities managed by provider
- Multitenant solutions

Issues

1. Browser based risks

- If the customer visits malicious website and browser becomes infected, the subsequent access to SaaS application might compromise the customer's data.
- To avoid such risks, the customer can use multiple browsers and dedicate a specific browser to access SaaS applications or can use virtual desktop while accessing the SaaS applications.

2. Network Dependence

- The SaaS application can be delivered only when network is continuously available.
- Also, network should be reliable, but the network reliability cannot be guaranteed either by cloud provider or by the customer.

3. Lack of portability between SaaS clouds

- Transferring workloads from one SaaS cloud to another is not so easy because workflow, business logics, user interfaces, support scripts can be provider specific.

Infrastructure-as-a-Service (IaaS)

- Infrastructure-as-a-Service (IaaS) is the delivery of computer infrastructure (typically a platform virtualization environment) as a service.
- IaaS leverages significant technology, services, and data center investments to deliver IT as a service to customers.
- Unlike traditional outsourcing, that requires extensive due diligence, negotiations and complex, lengthy contract , IaaS is centered arou nd a model of service delivery that provisions a predefined, standardized infrastructure specifically optimized for the customer's applications.

- IaaS includes the capability to provision processing, storage, networks, and other fundamental computing resources; the consumer is able to deploy and run arbitrary software, which can include operating systems and applications.
- The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems; storage, deployed applications, and possibly limited control of select networking components (e .g., host firewalls).
- Services offered by this paradigm include server hosting, web servers, storage, computing hardware, operating systems, virtual instances, load balancing, Int ernet access, and bandwidth provisioning.
- IaaS clouds often offer additional resources such as; images in a virtual machine image library, raw (block) and file-based storage, firewalls, IP addresses, virtual local area networks (VLANs) and software bundles

- Examples of IaaS providers include Amazon CloudFormation, Amazon EC2, Windows Azure Virtual Machines, DynDNS, Google Compute Engine, HP Cloud, Rackspace Cloud, ReadySpace Cloud Services, and Terremark.

- Rather than purchasing data center space, servers, software, network equipment, etc.,
  IaaS customers essentially rent those resources as a fully outsourced service.
- Usually, the service is billed on a monthly basis, just like a utility company bills customers.
- The customer is charged only for resources consumed.

The chief benefits of using this type of outsourced service include:

- Ready access to a preconfigured environment that is generally ITIL-based (The Information Technology Infrastructure Library [ITIL] is a customized framework of best practices designed to promote quality computing services in the IT sector.)
- Use of the latest technology for infrastructure equipment
- Secured, "sand-boxed" (protected and insulated) computing platforms that are us ually security monitored for breaches

- Reduced risk by having off-site resources maintained by third parties
- Ability to manage service-demand peaks and valleys
- Lower costs that allow expensing service costs instead of making capital investments
- Reduced time, cost, and complexity in adding new features or capabilities

Some of the most popular IaaS solutions are:

Compute as a service:

- One of the most ubiquitous IaaS offerings today, compute as a service provides compute capacity that includes servers, operating system access, firewalls, routers and load balancing on demand.
- These systems have management interfaces, and their capacity can be either shared or private.
- Depending on the provider and the options an enterprise chooses, compute as a service also can include automated patch management, management of infrastructure software, storage management, security management, dedicated customer support and customized SLAs.

Web hosting:

- Many organizations rely on their websites for marketing and revenue, and any glitch in operations can mean a loss of business.
- Moving a website to an IaaS based model ensures that the website won’t get bogged down during peak traffic times and that organizations won’t have to overpay for capacity to manage those traffic spikes.
- Loads will always be balanced, and uptime is guaranteed. SLAs help on this one.

Storage as Service:

- Storage as a Service is a business model in which a large company rents space in their storage infrastructure to a smaller company or individual.
- In the enterprise, SaaS vendors are targeting secondary storage applications by promoting SaaS as a convenient way to manage backups.
- The key advantage to SaaS in the enterprise is in cost savings -- in personnel, in hardware and in physical storage space.
- SaaS is also being promoted as a way for all businesses to mitigate risks in disaster recovery, provide long-term retention for records and enhance both business continuity and availability

Disaster recovery and backup as a service:

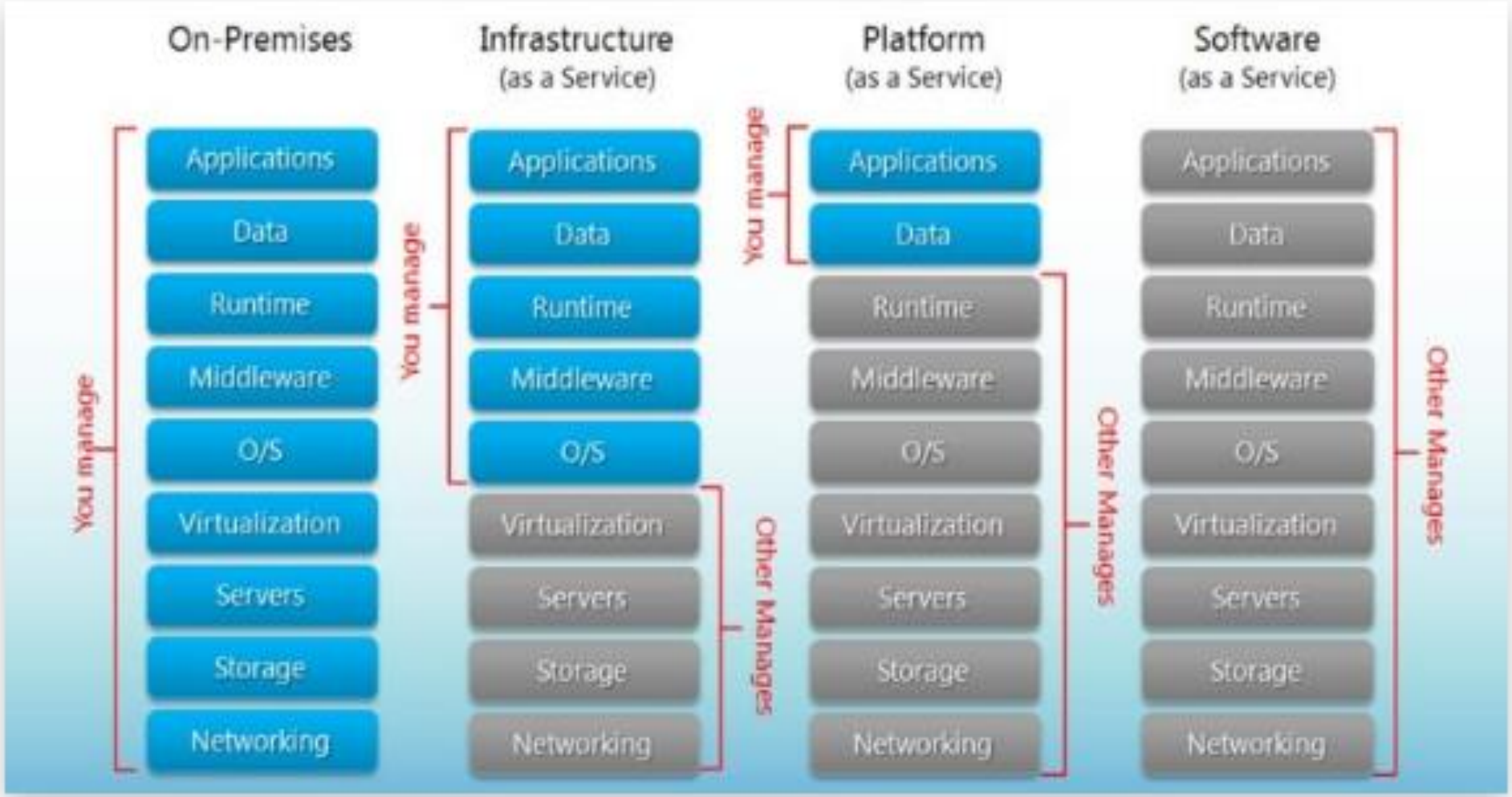
- The idea behind moving disaster recovery to the cloud is to ensure that organizations have uninterrupted access to data and applications, regardless of emergencies, such as power outages, natural disasters or system failures.
- These solutions always include redundancy and automatic failover to ensure ongoing access, reducing downtime to nearly zero.
- There are two basic options when it comes to disaster recovery as a service: backup and restore from the cloud and backup and restore to the cloud

Desktops as a service:

- DaaS is, in essence, an IaaS cloud created solely for hosting and serving virtual desktops.
- Essentially, it’s pay-as-you-go computing that allows enterprises to quickly provision, access, run and deactivate virtual desktop machines as needed.
- Organizations can choose to connect through a private network service instead of the public Internet. In most cases, the service provider offers storage for the virtual computers, ensures security and data protection, and controls the network bandwidth to ensure uptime.

Networking as a service:

- This is the newest entrant in the IaaS category.
- The idea is
  to offer networking resources on demand in order to support vir tual networks — resources such as firewalls, load balancing and WAN acceleration services.
- Simply put, NaaS provides unified connectivity across storage, networking and servers that changes to meet the demands of virtualized infrastructures.


On-Premises
Infrastructure
(as a Service)
Platform
(as a Service)
Software
(as a Service)
Applications
Data
Runtime
Middleware
O/S
Virtualization
Servers
Storage
Networking
You manage
Other Manages

Service Oriented Architecture(SOA)

- A Service-Oriented Architecture or SOA is a design pattern which is designed to build distributed systems that deliver services to other applications through the protocol.
- It is only a concept and not limited to any programming language or platform.
- SOA just makes it easier for software components over various networks to work with each other.
- In this architecture, services are provided to form applications, through a communication call over the internet.

There are two major roles within Service-oriented Architecture:

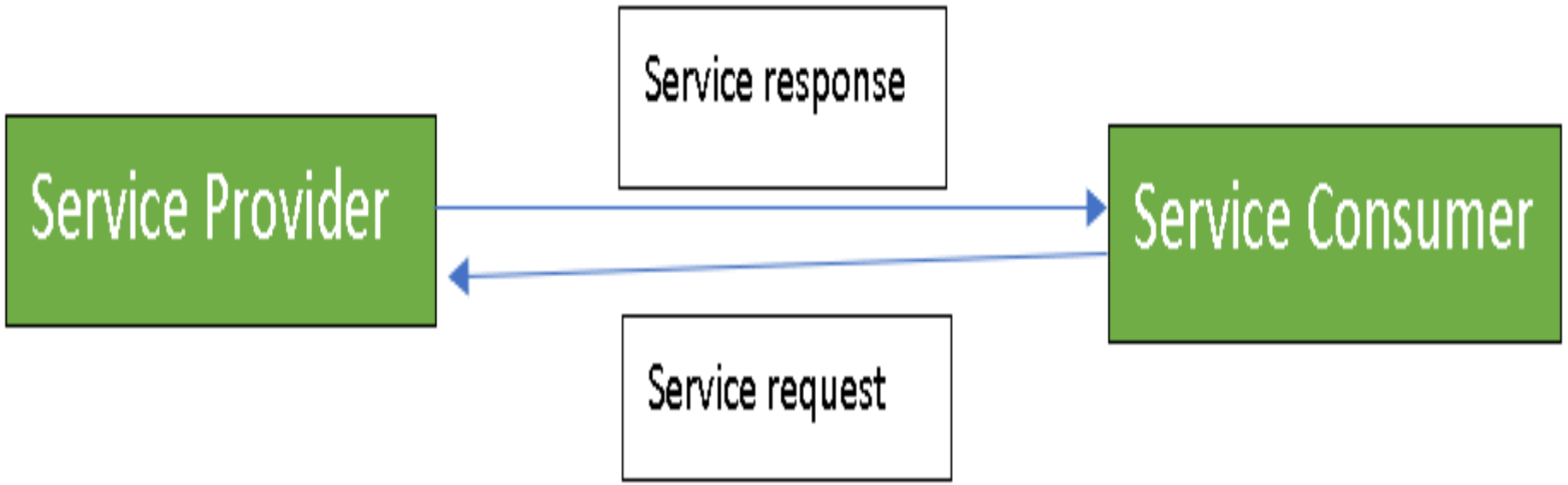
- **Service provider:**

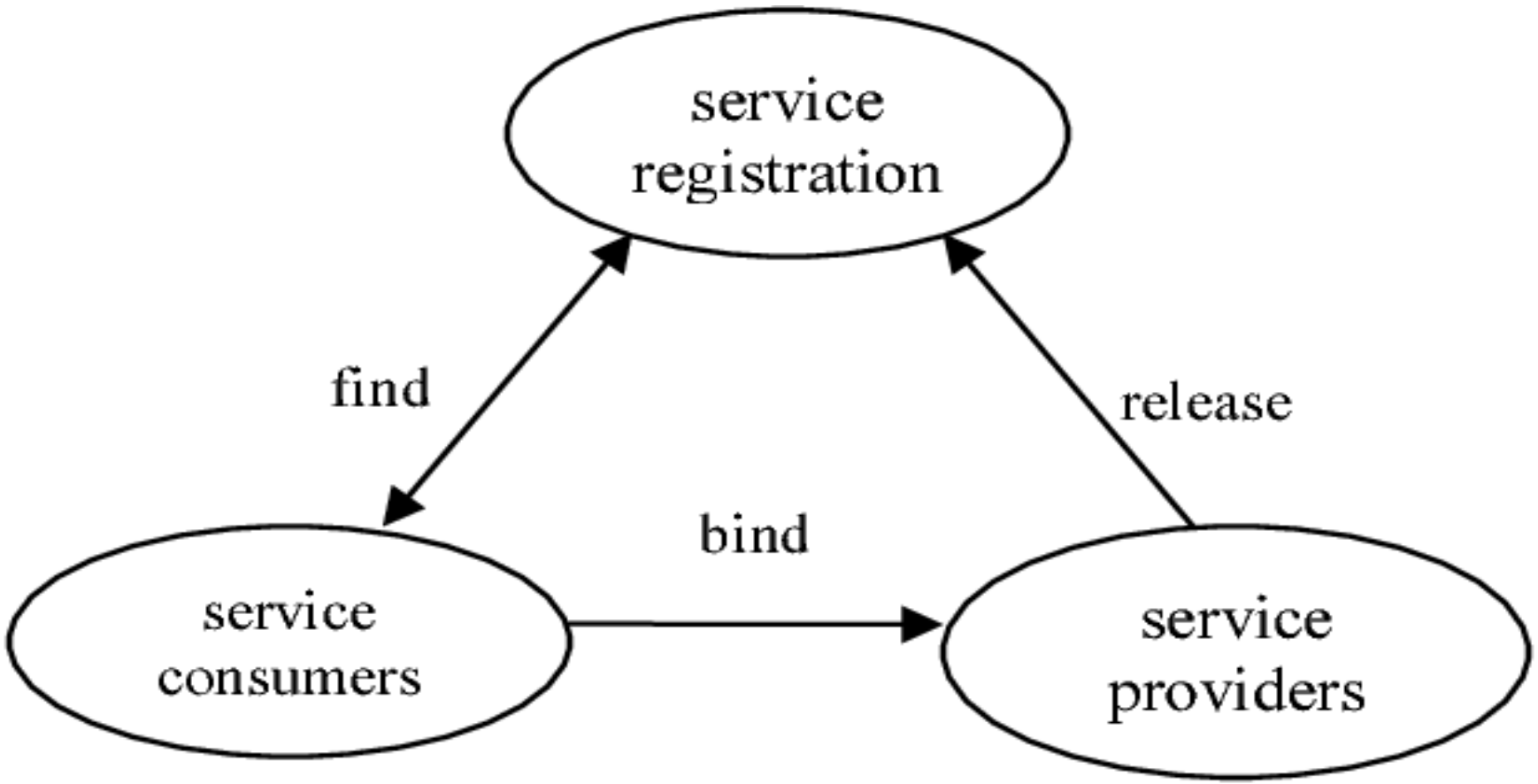
The service provider is the maintainer of the service and the organization that makes available one or more services for others to use.

To advertise services, the provider can publish them in a registry, together with a service contract that specifies the nature of the service, how to use it, the requirements for the service, and the fees charged.

- **Service consumer:**

The service consumer can locate the service metadata in the registry and develop the required client components to bind and use the service.


Service response
Service Provider
Service Consumer
Service request


service
registration
find
release
bind
service
consumers
service
providers

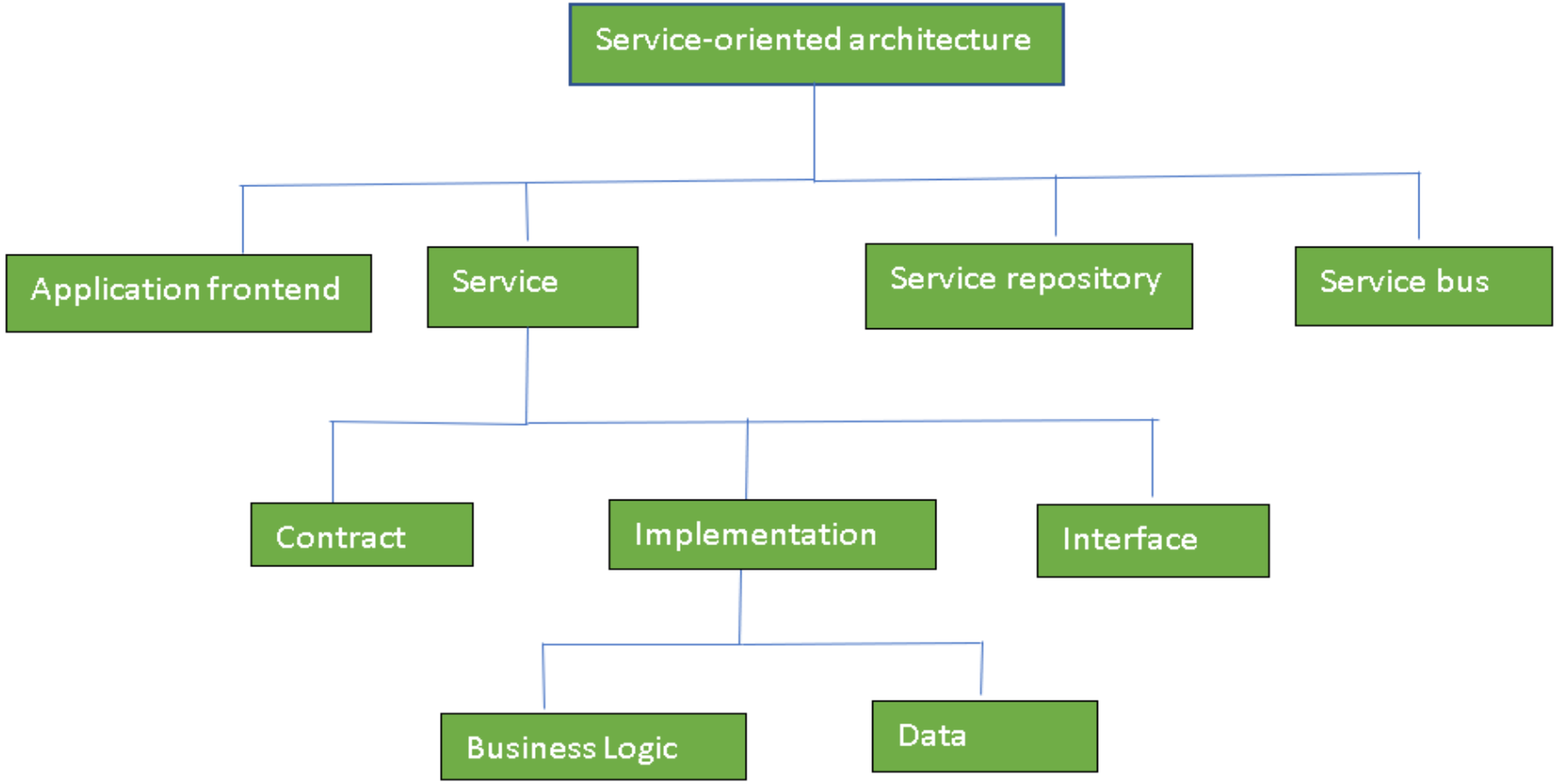
Principles of SOA

- **Standardized Service Contract** - Services adhere to a service description. A service must have some sort of description which describes what the service is about. This makes it easier for client applications to understand what the service does.
- **Loose Coupling** – Less dependency on each other. This is one of the main characteristics of web services which just states that there should be as less dependency as possible between the web services and the client invoking the web service. So, if the service functionality changes at any point in time, it should not break the client application or stop it from working.

- **Service Abstraction** - Services hide the logic they encapsulate from the outside world. The service should not expose how it executes its functionality; it should just tell the client application on what it does and not on how it does it.
- **Service Reusability** - Logic is divided into services with the intent of maximizing reuse. In any development company re-usability is a big topic because obviously one wouldn't want to spend time and effort building the same code again and again across multiple applications which require them. Hence, once the code for a web service is written it should have the ability work with various application types.

- **Service Autonomy** - Services should have control over the logic they encapsulate. The service knows everything on what functionality it offers and hence should also have complete control over the code it contains.
- **Service Discoverability** - Services can be discovered (usually in a service registry).
- **Service Composability** - Services break big problems into little problems. One should never embed all functionality of an application into one single service but instead, break the service down into modules each with a separate business functionality.

Components of SOA



How SOA works?

- With SOA, this software is organized as a set of software services.
- The services are supported by an infrastructure that, together with the services, improves information flow within the enterprise and between the enterprise and external enterprises.

SOA Architecture

- Service-oriented architecture is a style of software design.
- An architecture is categorized into two parts

1. functional aspects and
2. quality of service aspects.

Functional Aspects contains

- **Transport**: This component transports the service requests from the service consumer to the service provider and service responses from them to the service consumer.
- **Service Communication Protocol**: It allows the service provider and consumer to communicate with each other.
- **Service Description**: It explains the service and data required to invoke it.
- **Service**: It is an actual service.

- **Business Process:** This component represents the group of services called in a certain predefined sequence associated with the specific rules to meet the business demands.
- **Service Registry**: This registry contains the description of data that is used by service providers to publish their services.

Quality of service contains

- **Policy:** It is a set of protocols according to which service providers create and provide services to consumers.
- **Security:** It represents the set of protocols needed for the identification and authorization process.
- **Transaction:** It provides the surety of consistent results.
- **Management:** This component of SOA helps you define the set of attributes which are used to manage the services.

Advantage of SOA

- **Maintenance is Easy** – Editing and updating any service implemented under SOA architecture is easy. we don’t need to update our system. Service is maintained by a third party and any amendment in this service won’t influence our system.
- **Quality of Code Improved** – As services run independent of our system, they have their own variety of code, therefore, our code is prevented from redundancy. Also, our code becomes error free.
- **Platform Independence** – Services communicate with alternative applications through common language which implies it’s freelanced of the platform on that application is running. Services can provide API in different languages e.g., PHP, JavaScript, etc.

- **Scalable** – If any service is obtaining several users then it is often simply scalable by attaching additional servers. This will create service out there all time to the users.
- **Reliable** – Services square measure typically tiny size as compared to the full-fledged application. So, it’s easier to correct and check the freelance services.

. Same Directory Structure – Services have an equivalent directory structure so customers can access the service information from an equivalent directory on every occasion. If any service has modified its location, then additionally directory remains the same. This is very helpful for consumers.

7. Independent of Other Services – Services generated using SOA principles are independent of each other. So, services are often utilized by multiple applications at an equivalent time.

Disadvantages of SOA

- **High Bandwidth Server** -- As lots of information are sent and received, so need high bandwidth to manage those services
- **Extra Overload** – In SOA, all inputs square measures its validity before it’s sent to the service. If we are victimization of multiple services, then it’ll overload our system with further computation.
- **High Cost** – It is expensive in terms of human resource, development, and technology.

Combining SOA and Cloud Computing

- SOA is much more than a technological approach and methodology for creating IT systems.
- It’s also a business approach and methodology. Companies have used the principles of SOA to deepen the understanding between the business and IT and to help business adapt to change.
- Cloud services benefit the business by taking the best practices and business process focus of SOA to the next level.
- These benefits apply to both cloud service providers and cloud service users. Cloud service providers need to architect solutions by using a service-oriented approach to deliver services with the expected levels of elasticity and scalability.

- Companies that architect and govern business processes with reusable service-oriented components can more easily identify which components can be successfully moved to public and private clouds
- A Service-Oriented Architecture (SOA) is a software architecture for building business applications that implement business processes or services through a set of loosely coupled, black-box components orchestrated to deliver a well-defined level of service.

Characterizing SOA

- **SOA is a black-box component architecture.**
- The black box lets us reuse existing business applications; it simply adds a simple adapter to them.
- We don’t need to know every detail of what’s inside each component; SOA hides the complexity whenever possible.

- **SOA components are loosely coupled.**
- Software components are loosely coupled if they're designed to interact in a standardized way that minimizes dependencies.
- One loosely coupled component passes data to another component and makes a request; the second component carries out the request and, if necessary, passes data back to the first.
- Each component offers a small range of simple services to other components.
- A set of loosely coupled components does the same work that software components in tightly structured applications used to do, but with loose coupling we can combine and recombine the components in a bunch of ways. This makes a world of difference in the ability to make changes easily, accurately, and quickly.

- **SOA components are orchestrated to link through business processes to deliver a well-defined level of service.**
- SOA creates a simple arrangement of components that, together, deliver a very complex business service.
- Simultaneously, SOA must provide acceptable service levels. To that end, the components ensure a dependable service level.
- Service level is tied directly to the best practices of conducting business, commonly referred to as business process management (BPM) — BPM focuses on effective design of business process and SOA allows IT to align with business processes.

Cloud Computing Deployment Models

- Public
- Private
- Hybrid

Public Cloud

- Public clouds are available to the general public, and data are created and stored on third-party servers.
- Server infrastructure belongs to service providers that manage it and administer pool resources, which is why there is no need for user companies to buy and maintain their own hardware.
- Provider companies offer resources as a service both free of charge or on a pay-per-use basis via the Internet.
- Users can scale resources as required.

- The public cloud deployment model is the first choice for businesses with low privacy concerns.
- When it comes to popular public cloud deployment models, examples are Amazon Elastic Compute Cloud , Microsoft Azure, Google App Engine, IBM Cloud, Salesforce Heroku and others.

The Advantages of a Public Cloud

- **Hassle-free infrastructure management.** Having a third party running your cloud infrastructure is convenient: you do not need to develop and maintain your software because the service provider does it for you. In addition, the infrastructure setup and use are uncomplicated.
- **High scalability.** You can easily extend the cloud's capacity as your company requirements increase.
- **Reduced costs.** You pay only for the service you use, so there's no need to invest in hardware or software.
- **24/7 uptime.** The extensive network of your provider's servers ensures your infrastructure is constantly available and has improved operation time.

Disadvantage of Public Cloud

- **Compromised reliability.** That same server network is also meant to ensure against failure. But often enough, public clouds experience outages and malfunction, as in the case of the 2016 Salesforce CRM disruption that caused a storage collapse.
- **Data security and privacy issues give rise to concern.** Although access to data is easy, a public deployment model deprives users of knowing where their information is kept and who has access to it.
- **The lack of a tailor-made service.** Service providers have only standardized service options, which is why they often fail to satisfy more complex requirements.

Private Cloud

- There is little to no difference between a public and a private model from the technical point of view, as their architectures are very similar.
- However, as opposed to a public cloud that is available to the general public, only one specific company owns a private cloud.
- That is why it is also called an *internal* or *corporate* model.
- Compared to the public model, the private cloud provides wider opportunities for customizing the infrastructure to the company's requirements.

- A private model is especially suitable for companies that seek to safeguard their mission-critical operations or for businesses with constantly changing requirements.
- Multiple public cloud service providers, including Amazon, IBM, Cisco, Dell and Red Hat, also provide private solutions.

Benefits of a Private Cloud

- Tailor-made and flexible development and high scalability, which allows companies to customize their infrastructures in accordance with their requirements
- High security, privacy and reliability, as only authorized persons can access resources

Drawbacks of a Private Cloud

- The major drawback is its cost, as it requires considerable expense on hardware, software and staff training.
- That is why this secure and flexible computing deployment model is not the right choice for small companies.

Hybrid Cloud

- Hybrid cloud encompasses the best features of the abovementioned deployment models (public, private).
- It allows companies to mix and match the facets of both types that best suit their requirements.
- A company can balance its load by locating mission-critical workloads on a secure private cloud and deploying fewer sensitive ones to a public one.
- The hybrid cloud deployment model not only safeguards and controls strategically important assets but does so in a cost- and resource-effective way.
- In addition, this approach facilitates data and application portability.

Benefits of Hybrid Cloud

- Improved security and privacy
- Enhanced scalability and flexibility
- Reasonable price

Comparison

	Public	Private	Hybrid
Ease of setup and use	Easy	Requires IT proficiency	Requires IT proficiency
Data security and privacy	Low	High	High
Data control	Little to none	High	Comparatively high
Reliability	Low	High	High
Scalability and flexibility	High	High	High
Cost-effectiveness	The cheapest	Cost-intensive; the most expensive model	Cheaper than a private model but more costly than a public one
Demand for in-house hardware	No	Depends	Depends